

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	Sree B	Reg. No.:	192524023
Section / Batch:	AI and DS / 2026	Date:	01/08/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

1.A land survey application determines the boundary of a region using convex hull algorithms. a) Explain how brute-force convex hull identifies boundary points. b) Analyze its time complexity. c) Discuss why it is inefficient for large datasets.

2.A banking system stores daily transaction amounts in a list where new entries are appended randomly, causing disorder. The system needs to sort data periodically for reporting. a) Analyze how Selection Sort and Bubble Sort would work in this scenario. b) Compare the number of comparisons and swaps required by both methods. c) Recommend the most suitable sorting technique with proper justification.

Code of Conduct

I Sree B (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sree B

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. A land survey application determines the boundary of a region using convex hull algorithms. a) Explain how brute-force convex hull identifies boundary points. b) Analyze its time complexity. c) Discuss why it is inefficient for large datasets

The image shows two side-by-side Python IDE windows. The left window, titled 'IDLE Shell 3.13.7', displays the output of a program: 'Convex Hull Points: [(0, 3), (0, 0), (3, 3), (3, 0)]'. The right window, titled '1.py - C:/Users/SREE/OneDrive/Desktop/CSA0620 Design and Analysis of Algorithm/1.py (3.13.7)', contains the following Python code:

```
def orientation(p, q, r):
    return (q[0] - p[0]) * (r[1] - p[1]) - (q[1] - p[1]) * (r[0] - p[0])

def brute_force_convex_hull(points):
    n = len(points)
    hull = []

    for i in range(n):
        for j in range(i + 1, n):
            left = right = False

            for k in range(n):
                if k == i or k == j:
                    continue

                val = orientation(points[i], points[j], points[k])

                if val > 0:
                    left = True
                elif val < 0:
                    right = True

            if not (left and right):
                if points[i] not in hull:
                    hull.append(points[i])
                if points[j] not in hull:
                    hull.append(points[j])

    return hull

points = [(0,3), (2,2), (1,1), (2,1), (3,0), (0,0), (3,3)]

print("Convex Hull Points:")
print(brute_force_convex_hull(points))
```

2. A banking system stores daily transaction amounts in a list where new entries are appended randomly, causing disorder. The system needs to sort data periodically for reporting. a) Analyze how Selection Sort and Bubble Sort would work in this scenario. b) Compare the number of comparisons and swaps required by both methods. c) Recommend the most suitable sorting technique with proper justification.

The image shows two side-by-side Python IDE windows. The left window, titled 'IDLE Shell 3.13.7', displays the output of a program: 'Selection Sort: [1200, 2500, 3000, 4500, 7000]'. The right window, titled '2.py - C:/Users/SREE/OneDrive/Desktop/CSA0620 Design and Analysis of Algorithm/2.py (3.13.7)', contains the following Python code:

```
def selection_sort(arr):
    n = len(arr)

    for i in range(n):
        min_index = i

        for j in range(i + 1, n):
            if arr[j] < arr[min_index]:
                min_index = j

        arr[i], arr[min_index] = arr[min_index], arr[i]

    return arr

transactions = [4500, 1200, 7000, 2500, 3000]

print("Selection Sort:")
print(selection_sort(transactions))
```